

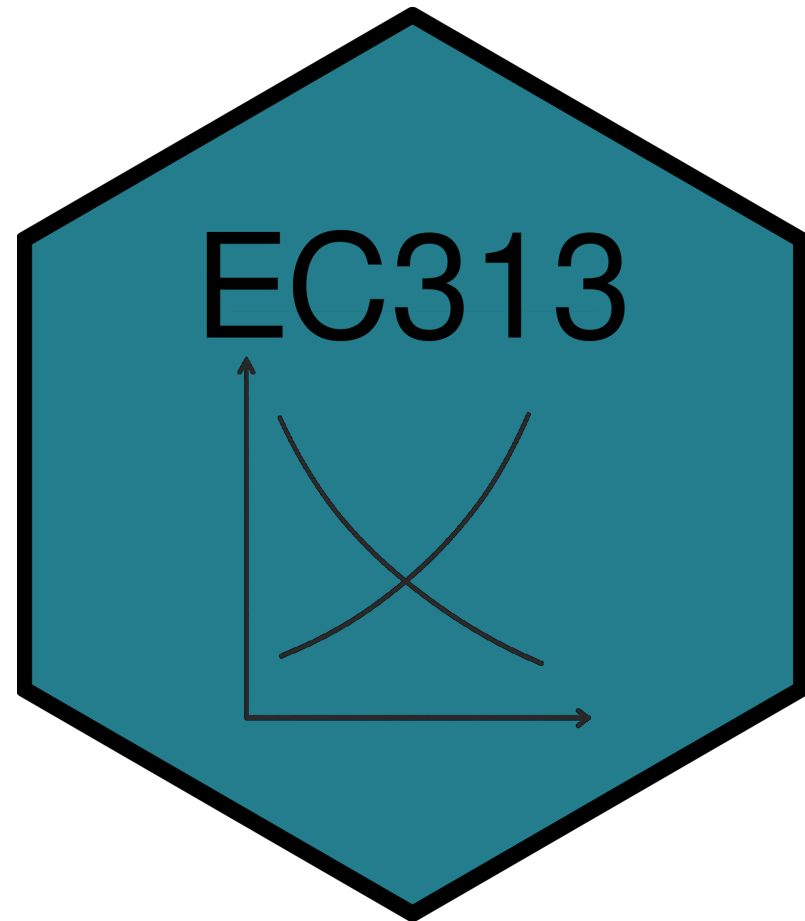
Personal Taxation and Behaviour

EC313 - Public Economics: Taxation

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Goals of This Section

In this section we will:

- Examine the effect of [income taxation](#) on [labour supply](#)
- Analyze how labour supply is related to tax revenues with the [Laffer Curve](#)
- Look at how taxation affects [saving behaviour](#)
- Discuss the preferential tax treatment of [primary residences](#) in Canada
- Examine taxation and [portfolio allocation](#)

Taxation and Labour Supply

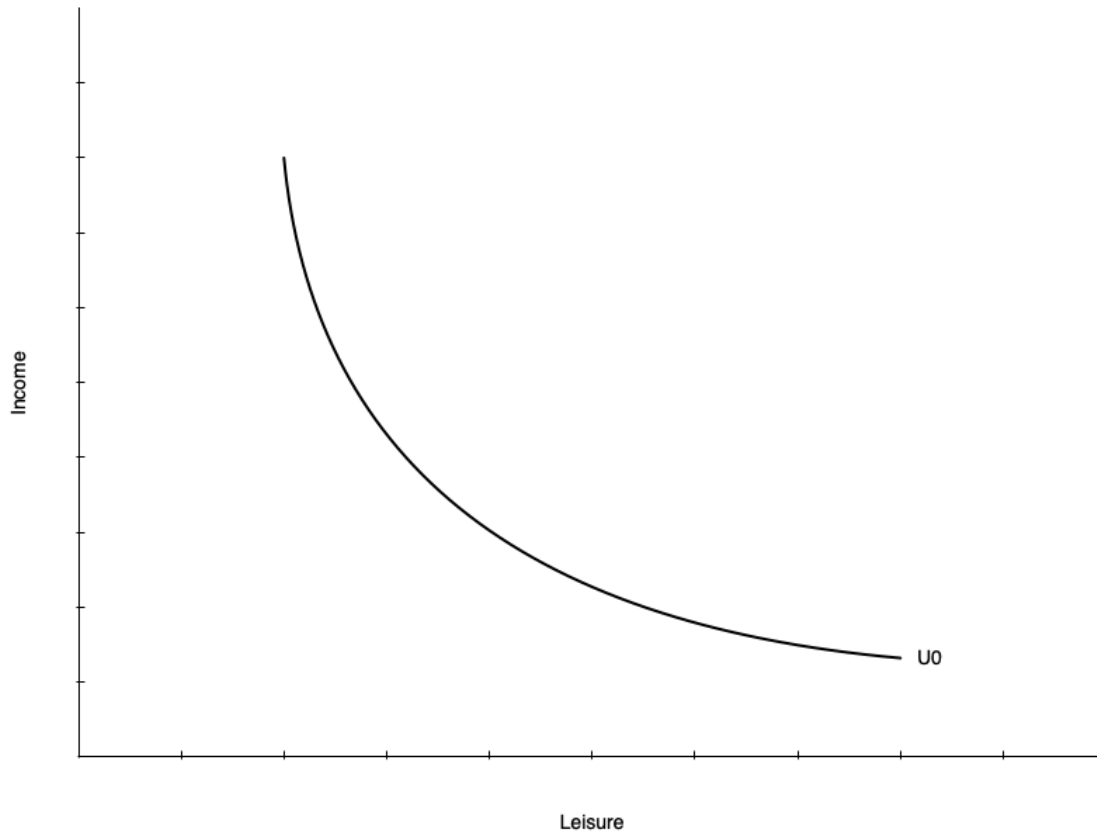
Introduction

- In this section we introduce a model we use throughout the course
- The model helps us understand how individuals make decisions about work
- Individuals choose between two goods
 - Income
 - Leisure
- Choices are constrained by time and wage rate
- There are a few wrinkles that make this market unique

Preferences

- Individuals have preferences over **Income (I)** and **leisure (L)**
 - People like having income to consume goods/services
 - Also like having non-work time
 - Includes fun activities but also non-work tasks like chores, education, etc.
- The tradeoffs people are willing to make between I and L are represented by **indifference curves**
 - At every point along the curve the individual is equally happy
 - Curves further from the origin represent higher levels of utility

Preferences



- On the left is a single indifference curve
- Any point represents equal utility (U_0)

! Marginal Rate of Substitution (MRS)

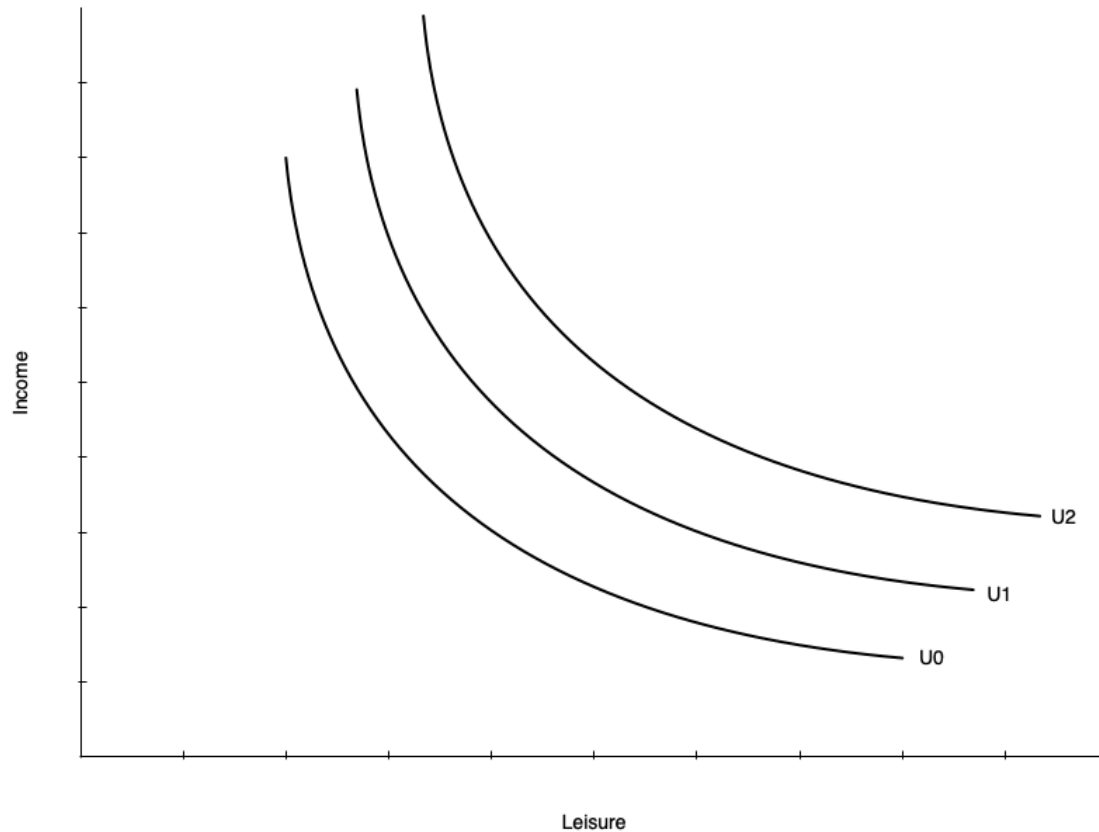
The rate at which a person is willing to trade income for leisure while remaining equally happy. The MRS is the slope of the indifference curve at a point.

- Individuals have **diminishing MRS**
 - Willing to give up less income for additional leisure the more leisure they already have
 - Slope gets shallower with more leisure

Preferences

- Indifference curves do not have to be smooth curves like the one shown
- Could be straight lines
 - Means the tradeoff between income and leisure is constant
 - Willingness to trade away leisure for income does not depend on leisure
- Could be “L” shaped
 - Income and leisure must be consumed in fixed proportions
- Mostly we will deal with smooth preferences

Preferences

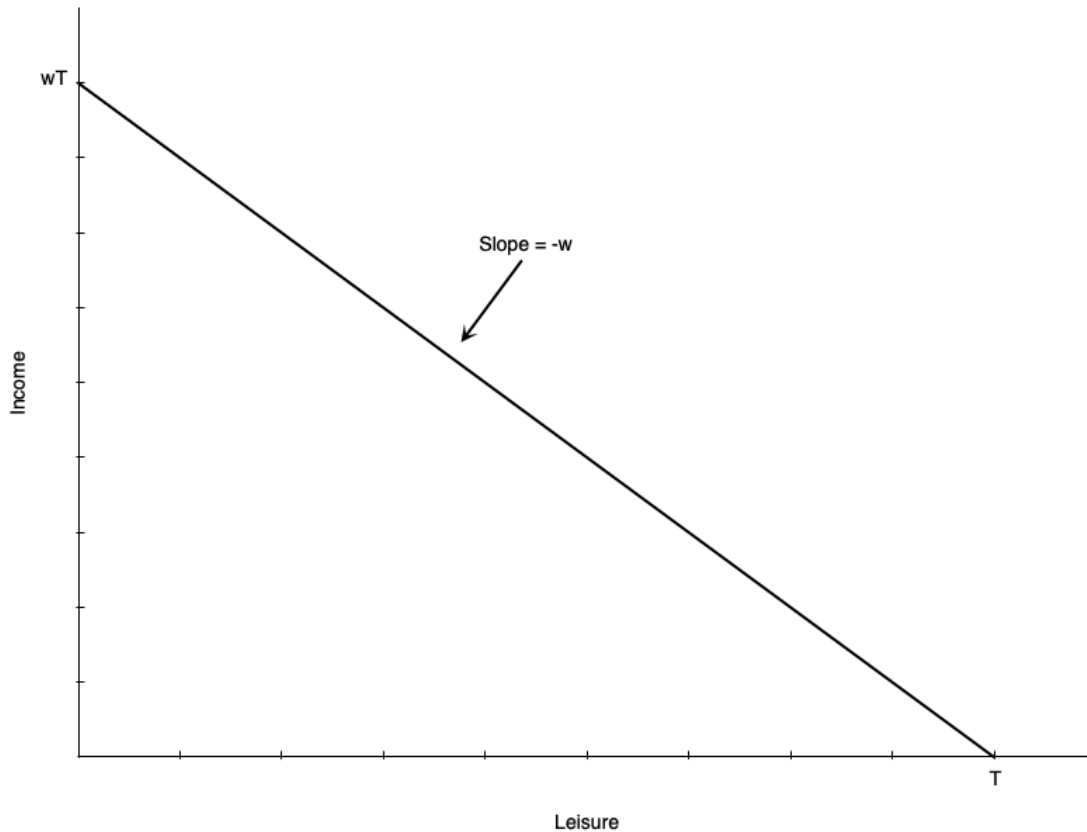


- Utility increases with curves further from the origin
 - $U_2 > U_1 > U_0$

Constraints

- Individuals do not choose income and leisure bundles on preference alone
- They are constrained by
 - The wage rate
 - Total number of hours available
- Budget constraint is also called
 - Potential income constraint
 - Full income constraint

Constraints



- Graph shows a linear budget constraint
- There are T total hours to allocate to work or leisure
- Each hour worked gets wage of w
- Person working T hours (no leisure) earns wT income
 - Called **full income**

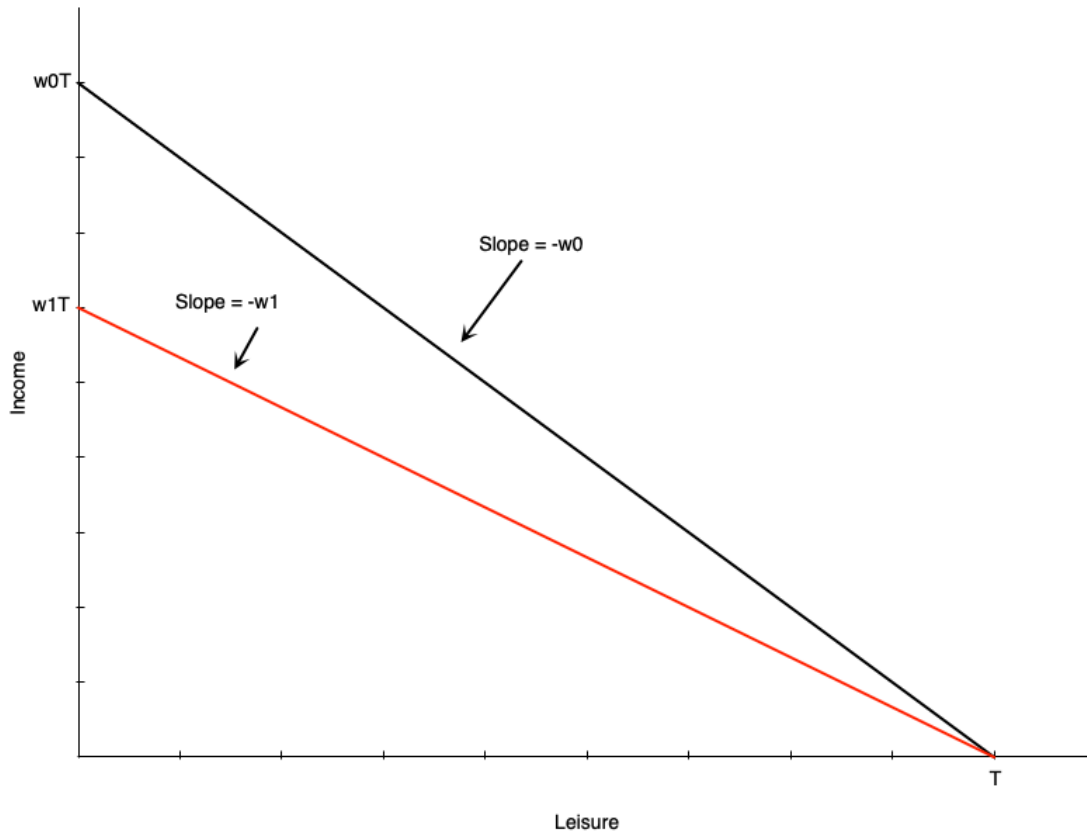
Constraints

- Mathematically, the budget constraint is

$$I = w(T - L)$$

- Income is equal to income from working ($w(T - L)$)
- The slope of the budget constraint is $-w$
 - The opportunity cost of leisure is the wage rate
 - Each hour of leisure means one less hour worked, which means w less income
- Someone who works zero hours has T leisure hours and no income
- Someone who works T hours has zero leisure hours and wT income

Constraints

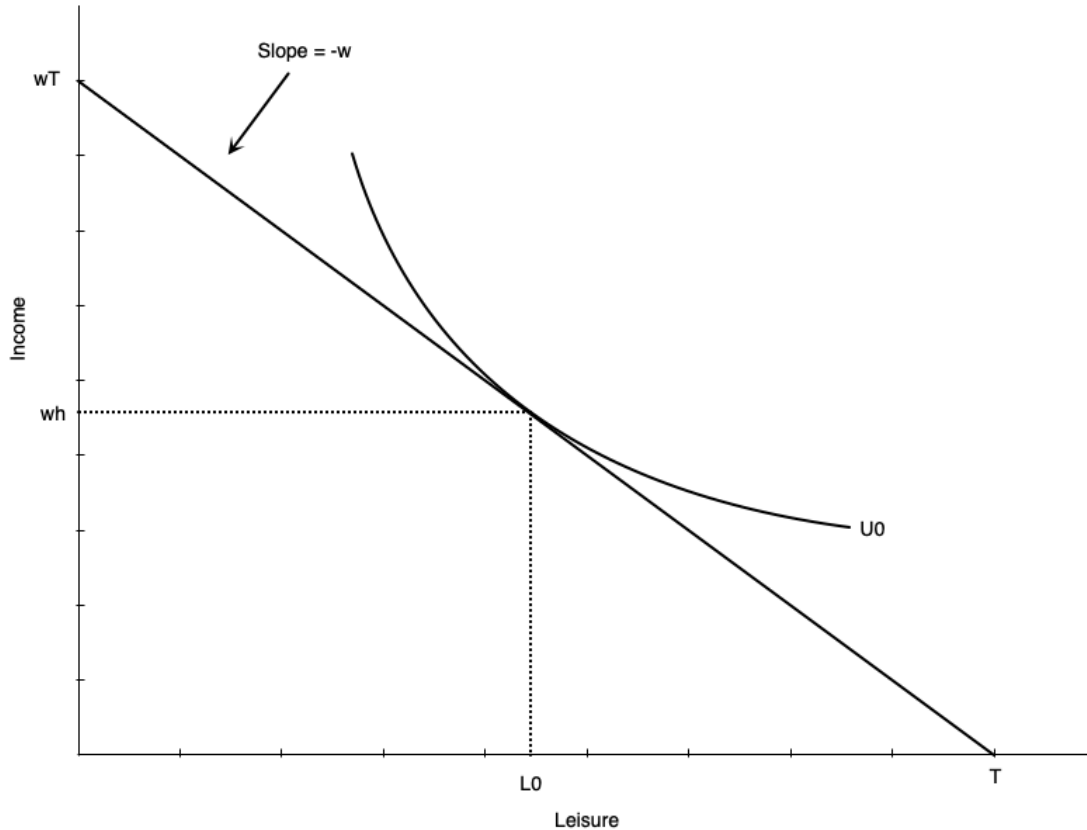


- Slope is the negative of the wage rate w
- A lower wage means shallower slope
 - If $w_1 < w_0$, the budget line swivels down
 - They intersect at leisure = T because there are no work hours
- Lower wage also means **full income** is lower
 - Working all hours leads to less total labour earnings

Consumer Optimum

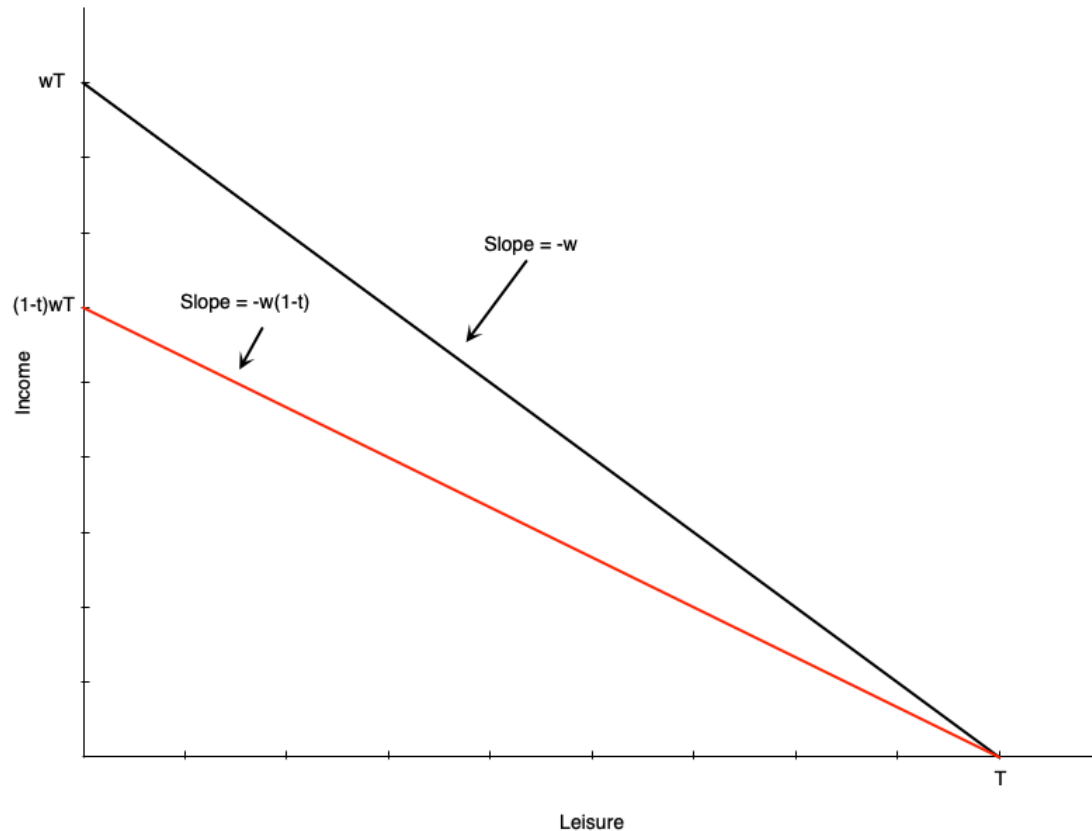
- Work hours are chosen using the combination of preferences and constraints
- The **consumer optimum** is the point that maximizes utility subject to the budget constraint
- The optimum occurs where the budget constraint is tangent to an indifference curve
- There are two possibilities for the optimum
 - Non-participation: optimum occurs at the end of the budget constraint (full leisure)
 - Participation: optimum occurs at an interior point (some leisure, some work)

Consumer Optimum



- Graph shows an optimum with participation
- Optimum is where indifference curve is tangent to budget constraint
- At the optimum, **MRS = wage rate**
 - Willingness to trade leisure for income (MRS) equals ability to trade them (wage)
- Optimal work hours are $h_0 = T - L_0$
- Optimal consumption is $I_0 = W_0 h_0$

Personal Income Tax



- The personal income tax affects the budget constraint
- Suppose a **proportional income tax** at rate t is introduced
- The after-tax wage is now $w(1 - t)$
- Tax causes the budget constraint to pivot inward
 - Slope is now $-w(1 - t)$
 - Full income is now $w(1 - t)T$

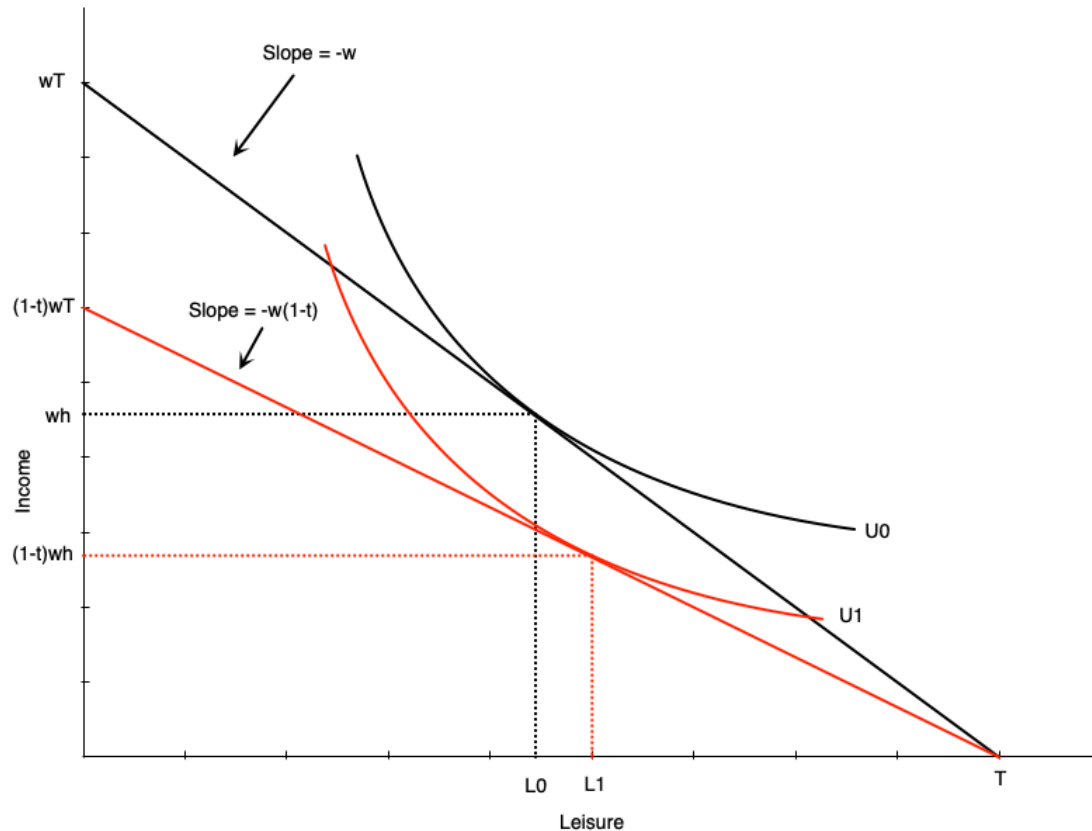
Personal Income Tax

- When the budget constraint changes, so does the optimum
- Suppose there is a proportional income tax at rate t
- There are two effects of the tax on labour supply

Two Effects of Income Tax on Labour Supply

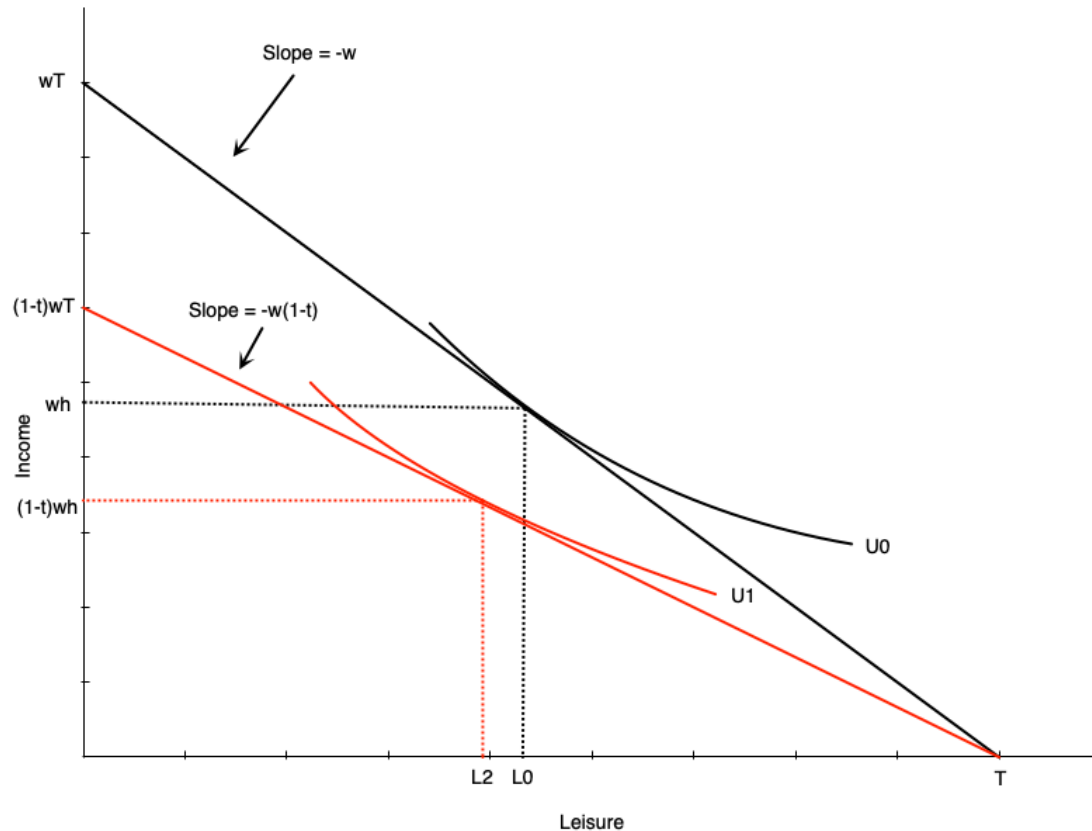
- **Substitution effect**: leisure is relatively cheaper, so people want more leisure (work less)
 - **Income effect**: people are poorer, so they want less leisure (work more)
-
- The effect on labour supply is therefore **ambiguous**
 - If substitution effect > income effect, labour supply **decreases**
 - If income effect > substitution effect, labour supply **increases**

Personal Income Tax



- Graph shows when **substitution effect** > **income effect**
 - Budget swivels down
 - Person optimizes at lower indifference curve
 - Leisure increases from L_0 to L_1
 - Work hours decrease from h_0 to h_1
- Happens when people are more willing to trade income for leisure

Personal Income Tax



- Graph shows when **income effect > substitution effect**
 - Budget swivels down
 - Person optimizes at lower indifference curve
 - Leisure decreases from L_0 to L_2
 - Work hours increase from h_0 to h_2
- Happens when people are less willing to trade income for leisure

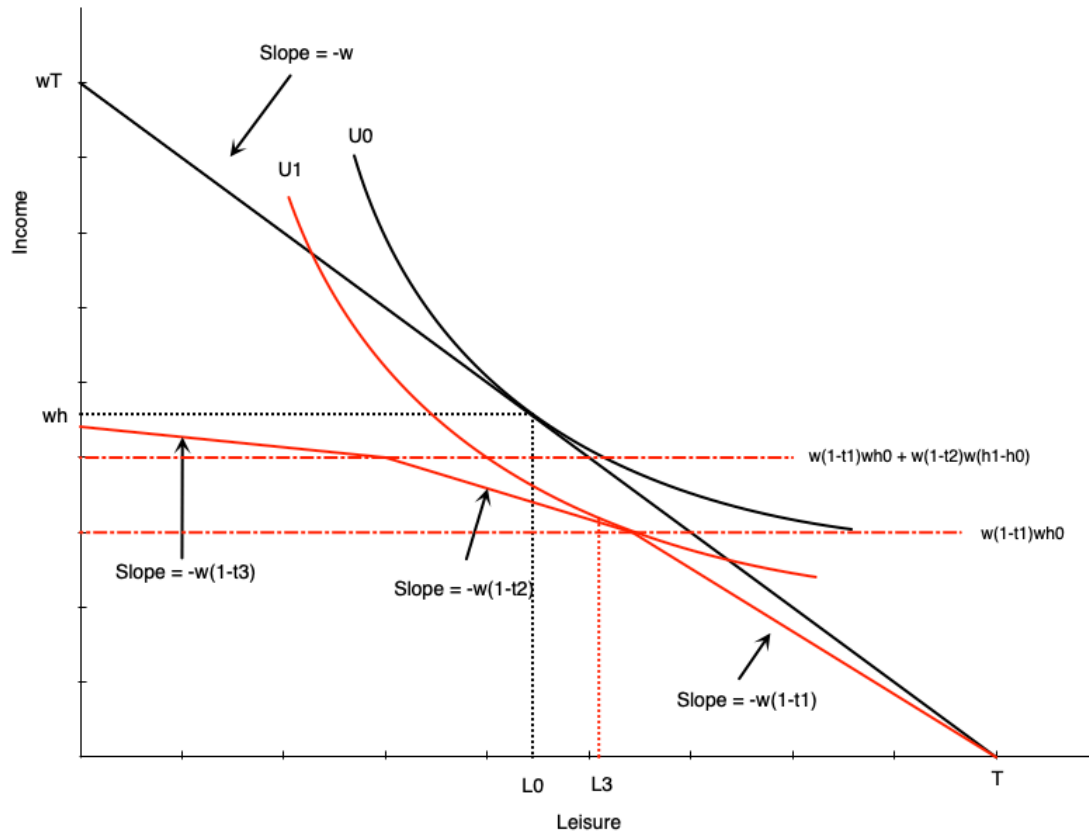
Think About It

A single parent working full-time faces a new income tax increase. Would you expect the substitution effect or the income effect to dominate their labour supply decision? How might your answer differ for a secondary earner in a high-income household?

Personal Income Tax

- In reality taxes are not flat
 - Canada has a **progressive income tax** system
 - Higher income is taxed at higher rates
- This implies that the budget constraint becomes kinked
 - As income goes up, the tax rate increases
 - When the tax rate changes, the slope changes
- The basic principles are the same
 - Substitution and income effects still apply
 - Labour supply effect is still ambiguous

Personal Income Tax



- Graph shows a progressive income tax with three brackets
- Tax rates are $t_3 > t_2 > t_1$
- Slope changes at each kink
 - First kink happens with work hours h_0
 - Second kink happens with work hours h_1
- Graph shows increased leisure, but effect is ambiguous

Evidence on Labour Supply

- The theory discusses effect of tax *holding all else equal**
- In statistical work, this is not easy to do
 - Many factors affect labour supply
 - Isolating the effect of taxes is difficult

Empirical Findings

- Elasticity of hours worked for working age men is between -0.2 and 0.2
 - For women it is between -0.2 and 0.5
-
- Means that a reduction in wage based on income tax has a **small effect** on work hours
 - Effect on **labour force participation** is larger

Caveats on Taxation and Labour Supply

Individual vs Group Effects

- We simplified the discussion by analyzing one specific type of person
- But tax changes affect many types of people differently
- A change from a flat to progressive tax might
 - Increase the tax rate for some people
 - Decrease the tax rate for others
- The way they react will be different and for policy analysis we need to consider the **group effects**

Human Capital

- Taxes can affect decisions about [education and training](#)
- We saw that taxes affect an individual's optimal labour supply at each wage
- If a tax increases the optimum hours at each wage, it would increase the desire to get education/training
 - This would make the worker more productive, and increase their wage
 - Could lead to further increases in supply
- The reverse would happen if tax decreases optimum hours at each wage

Total Compensation

- Wages are not the only part of compensation
 - There are non-wage benefits like health insurance, retirement contributions, etc.
- If taxes rise, firms may switch to **non-taxed forms of compensation**
- The effect of the tax may be muted if firms can do this easily

Expenditures

- Governments use tax revenues to fund public services
- That spending may itself affect labour supply
 - Spending on child care would **increase** supply
 - If spent on recreational facilities, parks, etc., may **decrease** supply
- Need to think about this from a **general equilibrium** lens

Labour Supply and Tax Revenues

Introduction

- As just noted, governments tax labour income to raise revenue
- The tax revenue collected from this exercise is the tax rate times the tax base
 - Tax base is the amount of income earned
 - Which in our model is $w \times h$
- So the tax revenue is $R = t \times w \times h$
- As we saw, if the substitution effect is larger than the income effect, higher tax rates reduce labour supply
- Means that as t increases, wh decreases

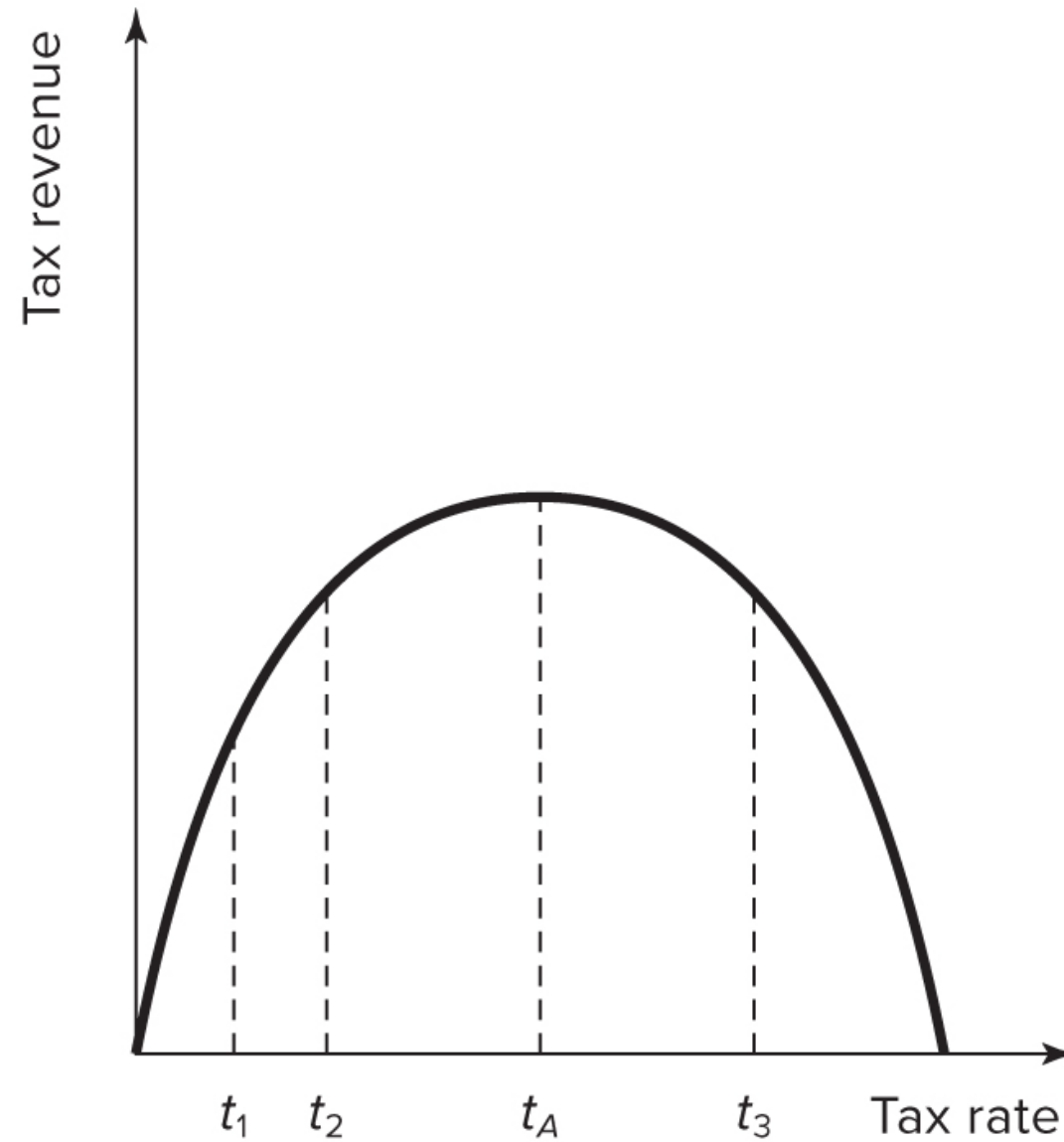
Laffer Curve

! The Laffer Curve

The relationship between tax rates and tax revenue. Tax revenue is zero at both 0% and 100% tax rates. Initially as tax rates rise, revenue rises, but at some point further increases in tax rates reduce revenue.

- The fact that higher tax rates may reduce labour supply has implications for tax revenue
- Initially as tax rates rise, revenue rises
- But at some point, further increases in tax rates **reduce revenue**
 - Because the reduction in labour supply is large enough that the tax base shrinks significantly

Laffer Curve



Laffer Curve

- The Laffer Curve suggests there is an **optimal tax rate** when it comes to revenues
 - At the peak of the curve where additional increases in tax rates do not increase revenue
- Also suggests that when tax rates are too high, **reducing them could increase revenue**
- However, determining the optimal tax rate is difficult in practice
 - Depends on how responsive labour supply is to tax changes
 - Empirical estimates suggest we are **not near the peak** of the Laffer Curve for labour income taxes in Canada

Think About It

Politicians sometimes argue that cutting tax rates will “pay for themselves” through higher economic activity and therefore higher revenue. Under what conditions would this claim be true, and why do most economists believe Canada is not currently in that situation?

Saving

Introduction

- In addition to labour supply, taxes affect [saving](#)
- Saving is important because it funds investment
 - Investment leads to capital accumulation
 - Capital accumulation leads to economic growth
- Taxes affect saving through reductions in the return
 - Interest earned through saving counts as income
 - That income is taxable
- In this section we will look at how taxes affect saving behaviour in a [life-cycle model](#)

Life-Cycle Model

- The [life-cycle model](#) assumes individuals plan their consumption and saving over their lifetime
- To keep things simple, we consider two time periods
 - Present and future
- Individuals have preferences for consumption in both periods
 - Represented by indifference curves
- They also have a budget constraint based on their income in both periods
 - Says that the present value of consumption equals the present value of income

Life-Cycle Model

- Algebraically, the budget constraint is

$$c_0 + \frac{c_1}{1+r} = I_0 + \frac{I_1}{1+r}$$

- Where
 - c_0 = consumption in present period
 - c_1 = consumption in future period
 - I_0 = income in present period
 - I_1 = income in future period
 - r = real interest rate

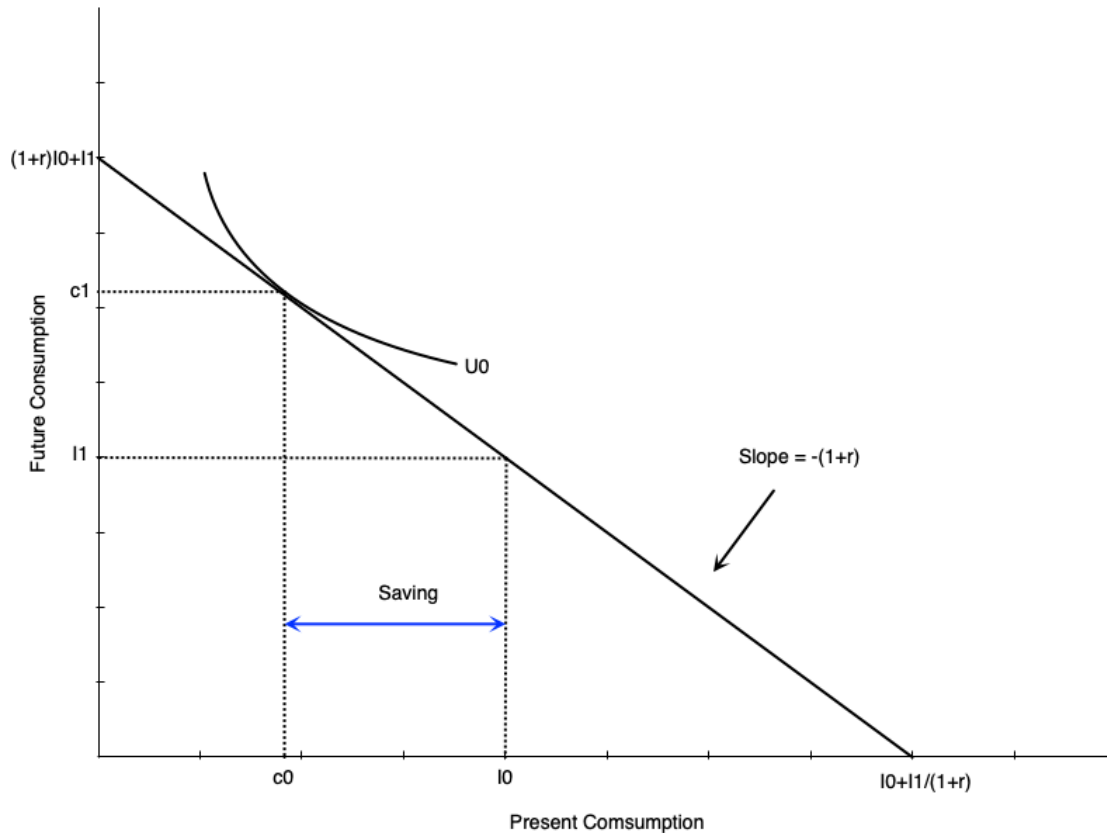
Life-Cycle Model

- You can plot this on a graph with c_0 on the x-axis and c_1 on the y-axis
- Rearrange to get c_1 on left side

$$c_1 = (1 + r)(I_0 - c_0) + I_1$$

- Slope of budget constraint is $-(1 + r)$
 - Consuming a dollar today means giving up $(1 + r)$ dollars tomorrow
- When a person consumes I_0 today, they consume I_1 tomorrow
 - The budget line passes through (I_0, I_1)

Life-Cycle Model



- Graph shows intertemporal budget constraint
- Slope is $-(1 + r)$
- Optimal consumption occurs where budget constraint is tangent to indifference curve
- In this case, the person saves in period 0 to consume more in period 1
 - **Saving** is the difference between income and consumption in period 0

Tax on Interest Income with Deductible Interest

- Now suppose there is a tax on interest income at rate t
- A person who saves \$1 in period 0 receives interest $r - rt$ in period 1
 - Because the interest earned is taxed they must pay rt on the interest
 - This lowers the return
- Means that foregoing \$1 in period 0 leads to consuming $1 + r - rt = 1 + (1 - t)r$ in period 1
 - The **after-tax interest rate** is $r(1 - t)$

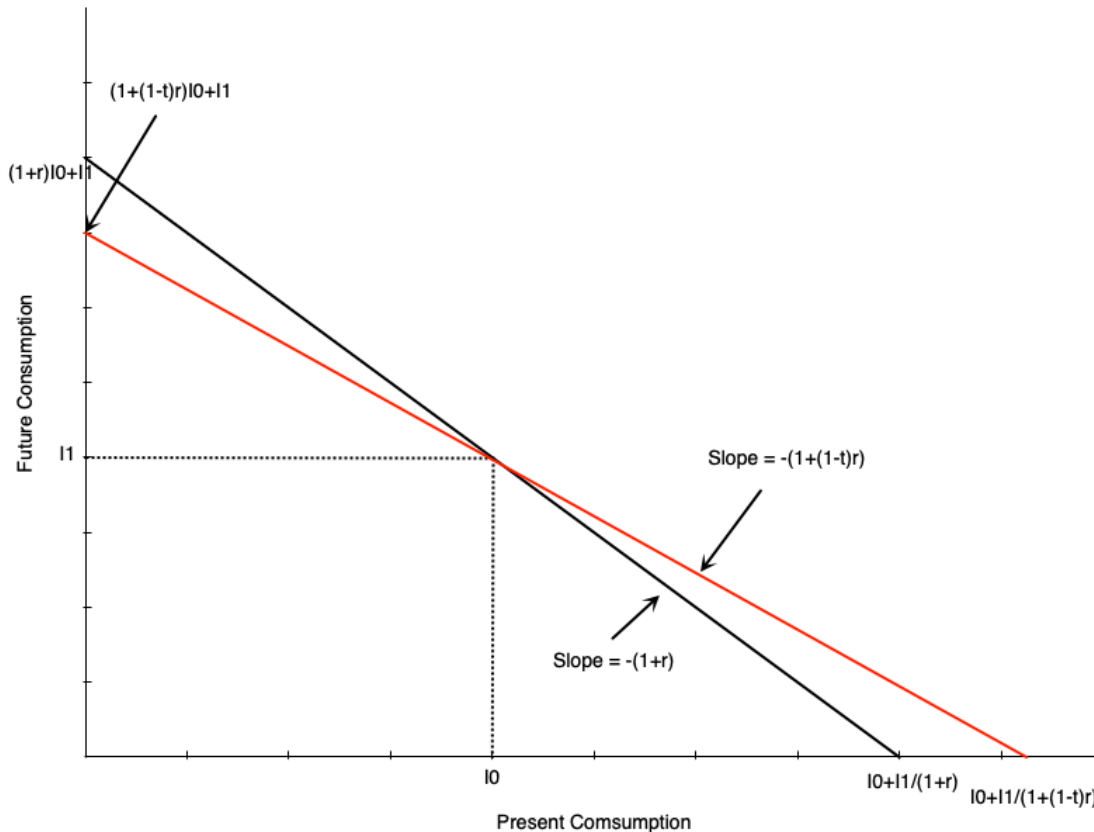
Tax on Interest Income with Deductible Interest

- A person who borrows \$1 in period 0 will need to pay interest $r - rt$ in period 1
 - They pay interest r
 - But because it is *tax deductible* they save rt on taxes
 - Means that total interest paid is $r - rt$
- Means that borrowing \$1 in period 0 leads to foregoing $1 + r - rt = 1 + (1 - t)r$ of consumption in period 1
 - The **after-tax interest rate** is $r(1 - t)$

Note

Whether saving or borrowing, the after-tax interest rate is the same when interest is deductible.

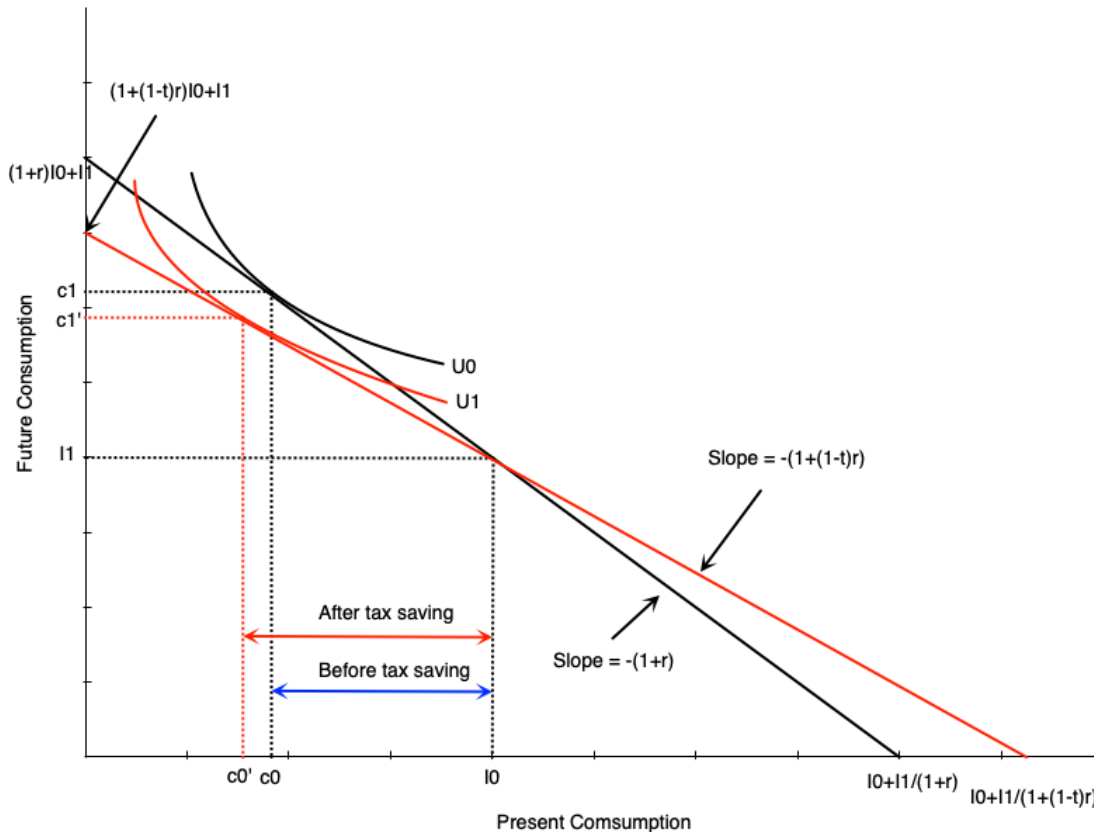
Tax on Interest Income with Deductible Interest



- Graph shows effect of tax on interest income on optimum
- Tax rotates budget constraint around (I_0, I_1)
 - Slope falls from $-(1 + r)$ to $-(1 + r(1 - t))$
 - Passes through (I_0, I_1) because no tax paid if no saving/borrowing
- Two effects of taxation
 - **Substitution effect**: opportunity cost of consumption today is lower, so consume more today (save less)
 - **Income effect**: lower real income, so consume less in both periods (save more)
- Total effect is **ambiguous**

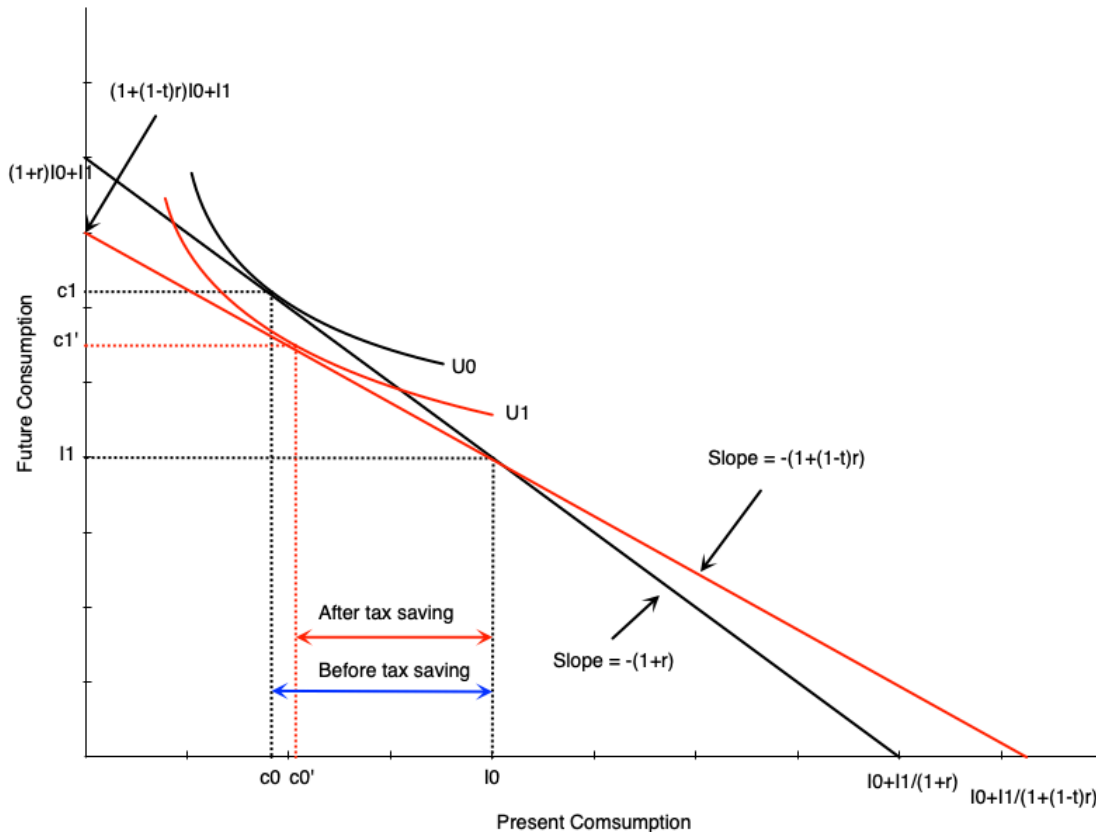
Tax on Interest Income with Deductible Interest

- Graph shows case when saving **increases**
 - Income effect > substitution effect
- Saving increases today
- But consumption in both periods falls



Tax on Interest Income with Deductible Interest

- Graph shows case when saving **decreases**
 - Substitution effect > income effect
- Saving falls today
- Consumption increases today, but falls in the future



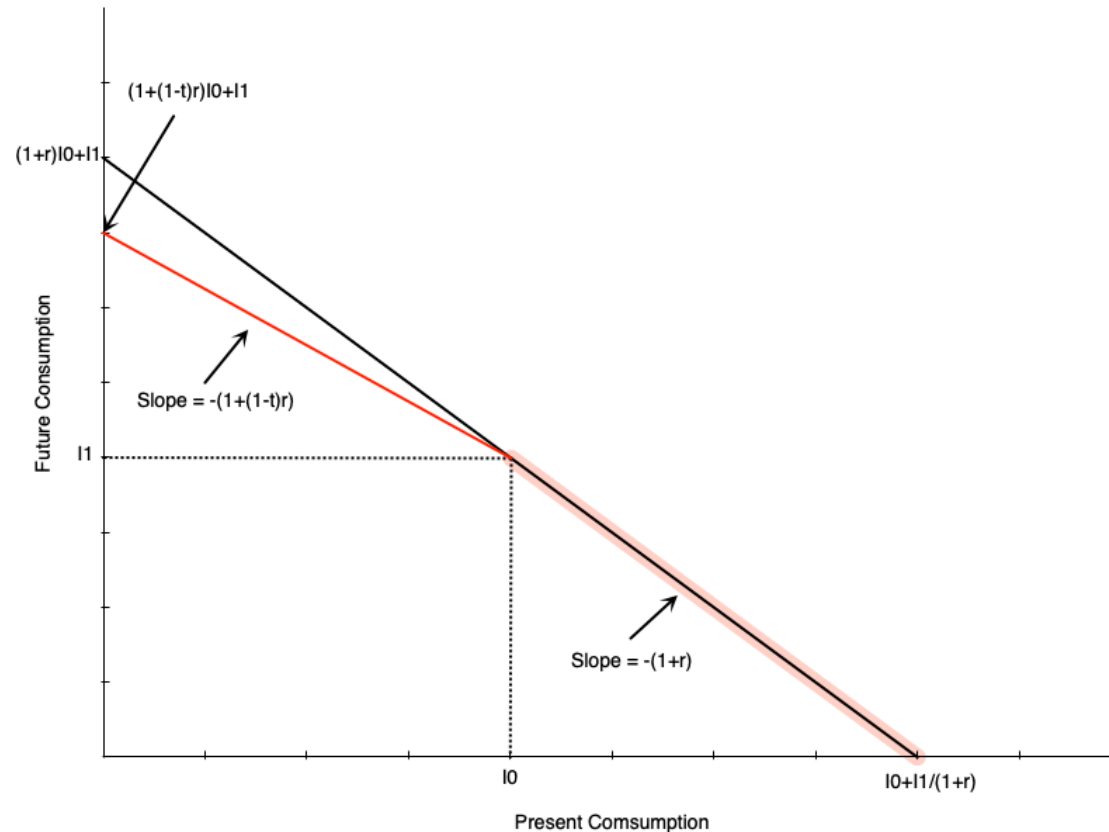
Think About It

Young workers early in their careers tend to earn less and may need to borrow, while older workers closer to retirement tend to save. How might removing the tax deductibility of interest payments affect these two groups differently, and what are the implications for intergenerational fairness?

Tax on Interest Income with Nondeductible Interest

- The deductibility of interest payments is important
 - It made the cost of borrowing the same as the return to saving
- Without deductibility, the after-tax interest rate for savers and borrowers differs
 - A saver who saves \$1 in period 0 receives interest $r - rt$ in period 1
 - A borrower who borrows \$1 in period 0 must pay interest r in period 1
- This creates a **kink** in the budget constraint at the point where consumption in both periods equals income

Tax on Interest Income with Nondeductible Interest



- Graph shows the budget constraint without deductible interest
- When the person saves ($c_0 < I_0$), the slope is $-(1 + r(1 - t))$
- When they borrow ($c_0 > I_0$), it is $-(1 + r)$
- The budget constraint is kinked at (I_0, I_1)
 - Pivots downward to the left of that point
 - Stays the same to the right

Tax on Interest Income with Nondeductible Interest

- For **savers** prior to the tax, the effects are the same as with deductible interest
 - **Substitution effect**: consume more today (save less)
 - **Income effect**: consume less in both periods (save more)
- For **borrowers** prior to the tax, there is **no effect**
 - The after-tax interest rate is unchanged at r
 - No substitution or income effect

Additional Considerations

- Models are intentionally simplifications of reality to help us understand key mechanisms
- The analysis is based on [real interest rates](#), but most saving is done with nominal rates
 - Inflation and indexation can affect the real return after tax
- There is more than one interest rate in the economy
 - Different types of saving have different tax treatments and returns
- Life cycle model may not be a very good representation of how people behave
 - Some people are not able to borrow the optimal level
 - Others are not good at saving

Evidence on Saving

- Researchers have analyzed the effect of taxation on saving

Key Finding

The interest elasticity of saving (responsiveness of saving to interest rate changes) is around 0.5, and varies by demographics including age.

- Results show that the response to taxation is **small**
- Generally a hard econometric analysis
 - Many factors affect saving behaviour
 - Isolating the effect of taxes is difficult

Savings Vehicles in Canada

- Canada has several ways to save
 - [Registered Pension Plans \(RPPs\)](#)
 - Tax deductible contributions, taxed withdrawals
 - [Registered Retirement Savings Plans \(RRSPs\)](#)
 - Tax deductible contributions, taxed withdrawals
- [Tax-Free Savings Accounts \(TFSAs\)](#)
 - Non-deductible contributions, tax-free withdrawals
- Other specialty ones like RESP, FHSA, etc.

Savings Vehicles in Canada

- One way to encourage saving is to increase limits on these vehicles
- Not clear that this would encourage new saving
 - People may just shift existing saving into these vehicles
 - Existence of other savings methods means people can substitute
- Empirical work gives mixed results
 - Tax deferred savings plans increased savings in the 1970s
 - More recently, increasing RRSP limit had little effect on total saving

Capital Taxation

- Taxing capital lowers the real return to investment
- As such, it may **reduce the incentive to invest**, and therefore capital accumulation and productivity
- Some argue that we should not tax capital income for this reason

Debate on Capital Taxation

- Empirical studies show saving not that responsive to returns
- In an open economy, domestic saving is used to fund domestic investment, but it may instead flow abroad
- For economic efficiency, should tax factors that create the least excess burden

Housing

Housing and Saving

- When we think of capital, it is usually machines
 - Used to produce goods in manufacturing
- Housing is also part of capital
 - Used to produce housing services
 - Generates income for some households

⚠ Preferential Tax Treatment of Housing in Canada

- No tax on **imputed rent** for owner-occupied housing
 - No **capital gains tax** on primary residences
-
- Leads to distortions in saving behaviour by encouraging investment in housing over other assets

Housing and Saving

- To see the effect, consider a household choosing between housing and other assets

Investing K in a stock:

- Earn R' dollars in money income
- Also earn C' in capital gains
- These are taxed, so overall return is $\frac{(1-t)(R'+C')}{K}$
- Tax treatment **favours housing**

Investing K in housing:

- Earn R dollars in imputed rent (an in-kind return)
- Also earn C in capital gains
- These are **not taxed**, so overall return is $\frac{R+C}{K}$

Housing and Saving

- What would the pre-tax return on the stock have to be to equalize after-tax returns?
 - Set after-tax returns equal:

$$\frac{(1 - t)(R' + C')}{K} = \frac{R + C}{K}$$

- Rearranging gives:

$$\frac{\frac{(R' + C')}{K}}{\frac{R + C}{K}} = \frac{1}{(1 - t)}$$

- Pre-tax stock return has to be $\frac{1}{(1-t)}$ times the housing return to be equally attractive
 - If $t = 0.25$, stock return has to be $\frac{1}{0.75} = 1.33$ times housing return



Housing and Saving

- Estimated foregone government revenue from non-taxation of capital gains on housing is around \$10 billion per year
- No estimates from non-taxation of imputed rent, but it would be higher
- **Preferential tax treatment** distorts saving behaviour
 - Encourages investment in housing over other assets
 - Funds may not flow to their most productive uses
- There are also some policies that affect rental prices of housing, which distort behaviour

Discussion

Canada's preferential tax treatment of housing has been in place for decades, and housing now represents a large share of household wealth. If the government were to remove the capital gains exemption on primary residences, who would benefit and who would be harmed? What political and economic obstacles make reforming housing taxation so difficult?

Changes in Tax Treatment of Housing

Should we tax imputed rent?

- Would **increase** government revenue
- But would be **difficult to administer**
- Could **reduce** homeownership rates
- Politically not feasible

Principal residence exemption

- An ongoing debate
- Since 2016 Canadians must declare a principal residence and its purchase price
- Also report the disposition (sale)

Portfolio Composition

Introduction

- People invest in different types of assets
 - Stocks, bonds, mutual funds, real estate, etc.
- These assets have different returns and risk profiles
- May also have different tax treatment
- By changing the real return, taxes will alter the [portfolio composition](#)

Portfolio Composition

- Imagine a simplified world with two assets
 - Cash: safe, but no return
 - Bond: positive return on average, but some risk of loss
- Prior to taxes, an investor decides on how much to hold of each
- Suppose then the government levies a tax on bond returns
 - Collects taxes if there is a positive return
 - Allows losses to be deducted from other income
- Two effects
 - **Average return**: because the tax reduces what the investor keeps, it reduces the return
 - **Risk**: because losses can be deducted, it reduces the risk of holding the bond

Portfolio Composition

- Lower return makes the bond **less attractive**
- Lower risk makes the bond **more attractive**
- Overall the effect is **ambiguous**
 - Without loss deduction, likely it would be less attractive to hold bonds
- Thus taxes affect portfolio composition, but effect can be complicated

Summary

Summary

- **Income taxation** affects labour supply through substitution and income effects
- The **Laffer Curve** shows the relationship between tax rates and tax revenues
- Taxes on interest income affect **saving behaviour** through substitution and income effects
- Preferential tax treatment of **housing** distorts saving behaviour towards housing
- Taxes affect **portfolio composition** by changing the return and risk of different assets

References

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